KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY, KUMASI

COLLEGE OF ENGINEERING

B.Sc. (Engineering), Second-Semester Examination, May, 2018

First (1st) Year

Agric, Aerospace, Biomedical, Chemical, Mechanical, Material and Petroleum Engineering

COE/EE 152: BASIC ELECTRONICS

TIME: 2 HRS

PROGRAM OF STUDY: ....

INDEX NO:………..........……………...

Instructions

i. Candidates are to indicate their Index Number and Program of Study in the spaces provided.
ii. Candidates are to submit ONLY the Scannable Sheet.
iii. Choose from the options lettered A to D, the most suitable answer to the question statements
iv. Shade the most suitable answer to the question statements on the Scannable Sheet provided
v. Candidates are to write the answer at the back of the Scannable Sheet if he/she thinks the right answer cannot be found in the options provided
vi. Each correct answer carries 0.75 a mark

CANDIDATES ARE ALLOWED TO TAKE THE QUESTION PAPERS AWAY

1. If a transistor is biased properly, it would work ………….. and not produce distortion in output signal
    a. Inefficiently
    b. Efficiently
    c. As a switch
    d. None of the above
2. Which of the following is preferred for biasing a typical transistor
    a. Battery
    b. Associating a circuit
    c. Associating a Diode circuit
    d. Both (a) and (b)
3. Biasing a transistor with a(n) ………..is an optimal biased condition
    a. Battery
    b. Associating a circuit
    c. Both of the above
    d. None of the above
4. Which of the following influences changes in the collector current of a transistor
    a. Temperature
    b. Transistor parameters
    c. Both of the above
    d. None of the above
5. The ……………. of operating point stable is key for operation of a transistor
    a. Amplification
    b. Maintenance
    c. Stabilisation
    d. Switching
6. The selection of a proper quiescent point generally depends on the following factors:
    a. The frequency of the signal to be handled by the amplifier and distortion level in signal
    b. The load to which the amplifier is to work for a corresponding loud speaker supply voltage
    c. Both of the above
    d. None of the above
7. The ………..of a transistor amplifier shifts mainly with changes in temperature
    a. Operating point
    b. Emitter
    c. Collector
    d. Base
8. …….., ……… and …….are functions of temperature
    a. β, I_{CO} and V_{EE}
    b. β, I_{CO} and V_{CC}
    c. β, I_{CO} and V_{CE}
    d. β, I_{CO} and V_{BE}

9. For a transistor circuit to amplify it must be properly biased with………..
    a. ac currents
    b. ac voltages
    c. dc voltages
    d. dc currents
10. The dc operating point between saturation and cutoff is called the ……….
    a. Action point
    b. Q-point.
    c. Amplification point
    d. None of the above
11. The goal is to set the ……….such that it does go into saturation or cutoff when an AC signal is applied.
    a. Turn-on value
    b. Q-point
    c. Temperature
    d. None of the above
12. Which of the following is **NOT** a requirement of biasing networks/circuits
    a. Ensuring proper zero signal collector current.
    b. Ensuring V_{CE} not falling below 0.3V for Ge transistor and 0.7V for Silicon transistor at any instant.
    c. Ensuring Stabilization of operating point. (zero signal I_C and V_{CE})
    d. None of the above
13. The DC current gain (β) of a transistor remains constant at 200 at 75^{o}C and the base current (I_B) at 20.28μA while the collector current (I_C) changes from 2.028mA to 3.04mA, evaluate the thermal stability factor (S) of the operating point of this transistor.
    a. 35
    b. 25
    c. 50
    d. None of the above
14. Which of the following is not a condition for the solution in **question 13** to remain always true?
    a. I_B and alpha must remain constant
    b. I_B and beta must vary
    c. Alpha and beta must remain constant
    d. (a), (b) and (c)
15. Which of the following is the desired range of values for the thermal stability factor (S) of the operating point of a transistor?
    a. 1 to 50
    b. 1 to 25
    c. 1 to 75
    d. Has no range

16. Which of the following is the ideal value(s) for the thermal stability factor (S) of the operating point of a transistor?
    a. 1
    b. 25
    c. 50
    d. 15

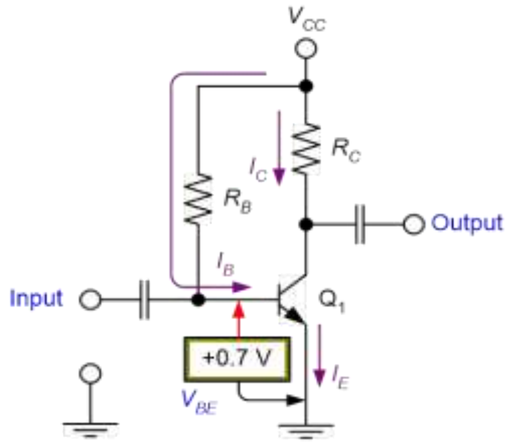

Figure 1.0

Ing Patapaa wants to design a circuit to be used to control the main door at his studio. As a guru in electronics he designed the circuit shown in Figure 1.0. After designing the circuit, he measures the values of h_{FE} = 100 when temperature (T) = 25 °C and h_{FE} = 150 when T = 100 °C. Given that R_B = 300kΩ, R_C = 1.3kΩ, and V_{CC} = +12V

Use figure 1.0 and the information below it to answer questions from 17 to 28

17. Determine the value(s) of the base current at both of these temperatures
    a. 377.6μA
    b. 3.767μA
    c. 376.7μA
    d. 37.67μA
18. Determine the Q-point value I_C at the temperature (T) = 25 °C.
    a. 377.6mA
    b. 3.767mA
    c. 376.7mA
    d. 37.67mA
19. Determine the Q-point value I_C at the temperature (T) = 100 °C.
    a. 57.67mA
    b. 5.767mA
    c. 5.650mA
    d. 56.50mA
20. Determine the Q-point value V_{CE} at the temperature (T) = 25 °C.
    a. 7.1030V
    b. 6.1030V
    c. 4.6550V
    d. 5.6550V

21. Determine the Q-point value V_{CE} at the temperature (T) = 100 °C.
    a. 7.1030V
    b. 6.1030V
    c. 4.6550V
    d. 5.6550V
22. Determine the value of H_{FE} considering both temperatures (T=25°C and 100 °C)
    a. 122.13
    b. 122.10
    c. 122.47
    d. 122.55
23. Determine the value of I_C using the value of H_{FE} from question 22
    a. 46.14mA
    b. 4.614mA
    c. 4.674mA
    d. 46.74mA
24. Determine the value of V_{CE} using the values from question 23
    a. 6.1030V
    b. 6.0018V
    c. 6.1800V
    d. 6.3010V
25. Determine the value(s) of I_C and V_{CE} at Saturation and Cut-off respectively at the average value of H_{FE}
    a. 9.230 mA and 12.00V
    b. 8.230 mA and 6.93V
    c. 7.230 mA and 12.00V
    d. 6.230 mA and 7.94V
26. Which of the following is an advantage and a disadvantage of the biasing technique employed in figure 1.0
    a. Circuit is Complex and Q-point shift with change in temperature
    b. Good Amplification and Q-point shift with change in temperature
    c. High Stability and Q-point shift with change in temperature
    d. None of the above
27. The values of I_C and V_{CE} at T = 25 °C compared to the values of I_C and V_{CE} at Saturation and Cut-off respectively it can be said that………
    a. Average bias circuit
    b. Mid-point bias circuit
    c. Maximum bias circuit
    d. None of the above
28. Which of the following is a(n) application for the biasing type shown in figure 1.0
    a. Switching
    b. Trimming
    c. Regulating
    d. Amplification

The diodes used in a bridge rectifier circuit have the following parameters:

$V_T = 0V$, $r_f = 20\Omega$, $R_R = \infty$, $V_Z =$ very large

The secondary winding of the transformer has a resistance $R_S = 20\Omega$. A load of 990Ω is connected across its output. If the voltage across the secondary is $v_s = 40 \sin wt$ volts,

Use the above information to solve questions 29 to 37

29. Determine the PIV rating of the diodes
    a. 40V
    b. 20V
    c. 60V
    d. 10V
    e. None of the above
30. Determine the DC current through the load considering the voltage drop across the Diodes
    a. 0.06024A
    b. 0.24720A
    c. 0.02472A
    d. 0.00247A
    e. None of the above
31. Determine the DC Voltage on the load considering the voltage drop across the Diodes
    a. 12.328V
    b. 12.238V
    c. 12.823V
    d. 12.338V
    a. None of the above
32. Determine the DC power supplied to the load
    a. 0.3605W
    b. 0.3280W
    c. 0.3840W
    d. 0.3025W
    e. None of the above
33. What is the conversion efficiency of this rectifier
    a. 0.7637pu
    b. 0.7285pu
    c. 0.7640pu
    d. 0.7437pu
    e. None of the above
34. What is the peak value of the AC current?
    a. 0.2020A
    b. 0.2128A
    c. 0.78408A
    d. 0.20406A
    e. None of the above

35. What is the AC power supplied to the whole circuit?
    a. 0.2020W
    b. 0.2128W
    c. 0.78408W
    d. 0.20406W
    e. None of the above
36. What is the RMS value of the current supplied to the circuit?
    a. 0.1428A
    b. 0.1248A
    c. 0.2428A
    d. 0.20406A
    e. None of the above
37. Which of the following will improve the efficiency of the rectifier
    a. Employing a capacitor for filtering with practically zero internal resistance
    b. Employing a zener diode for regulation with ideal internal resistance
    c. Employing a diode with practically zero internal resistance
    d. Employing a transformer with practically zero internal resistance
    e. None of the above

Preamble for questions (38 to 46). Ing Shatta Wale having studied basic electronics for a semester wanted to design a voltage regulator which will handle a load of 4W connected across a 40V zener regulator. The zener current can vary from 50mA to 100mA whilst maintaining a constant voltage output. Considering the series resistor to be 1kΩ

38. Determine the current flowing through the load connected to the regulator
    a. 150mA
    b. 100mA
    c. 240mA
    d. 190mA
    e. None of the above
39. What would be the value of current flowing through the series resistor when minimum current is flowing through the zener diode?
    a. 150mA
    b. 100mA
    c. 240mA
    d. 190mA
    e. None of the above
40. What would be the value of current flowing through the series resistor when maximum current is flowing through the zener diode?
    a. 150mA
    b. 100mA
    c. 200mA
    d. 190mA
    e. None of the above

41. What would be the voltage drop across the series resistor when the minimum current is flowing through the zener diode?
    a. 150V
    b. 200V
    c. 240V
    d. 190V
    e. None of the above
42. What would be the voltage drop across the series resistor when the maximum current is flowing through the zener diode?
    a. 150V
    b. 200V
    c. 240V
    d. 190V
    e. None of the above
43. Find the minimum input voltage if the series resistor is 1kΩ.
    a. 150V
    b. 200V
    c. 240V
    d. 190V
    e. None of the above
44. Find the maximum input voltage if the series resistor is 1kΩ.
    a. 150V
    b. 200V
    c. 240V
    d. 190V
    e. None of the above
45. What is the maximum power the zener diode can withstand?
    a. 4W
    b. 3W
    c. 3.5W
    d. 3.9W
    e. None of the above
46. How would you determine if the zener diode used in the circuit is ON or OFF?
    a. Comparing the voltage ratings of the zener diode to the voltage drop across the load.
    b. Comparing the voltage ratings of the zener diode to the voltage drop across the filter capacitor.
    c. Comparing the voltage ratings of the zener diode to the voltage drop across the zener diode.
    d. Comparing the voltage ratings of the zener diode to the voltage drop across the transformer.
    e. None of the above

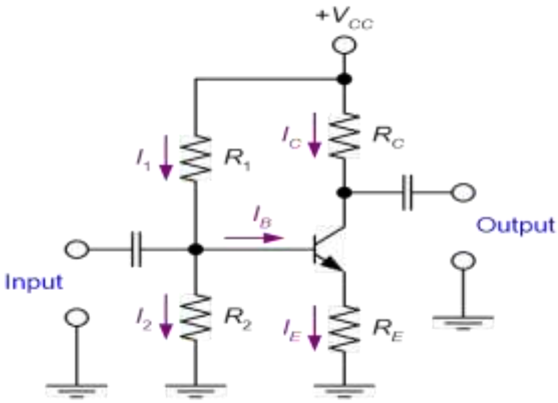

Figure 1.10

DJ Klogo is designing an amplifier so that he can produce the best mix when he has a gig. He employs the transistor circuit shown in ***Figure 1.10***. The specifications from the data sheet of the transistor has h_{FE} = 100 when at room temperature, T = 25 °C. Given that R_1 = 20kΩ, R_2 = 5kΩ, R_E = 1.3kΩ, R_C = 3kΩ, and V_{CC} = +12V. Assuming the base current (I_B) is negligible.

Use figure 1.10 and the information below it to answer questions from 47 to 58

47. Determine the base voltage (V_B) on the transistor.
    a. 2.12V
    b. 2.00V
    c. 2.21V
    d. 2.20V
    e. None of the above
48. Determine the value of the current (I_2) flowing through resistor (R_2)
    a. 3.000μA
    b. 400.0μA
    c. 4.000μA
    d. 300.0μA
    e. None of the above
49. Determine the value of the voltage (V_E) on the emitter of the transistor
    a. 1.12V
    b. 1.20V
    c. 1.21V
    d. 1.30V
    e. None of the above
50. Determine the value of the current (I_E) flowing through the emitter
    a. 1.12mA
    b. 1.00mA
    c. 1.21mA
    d. 1.20mA
    e. None of the above

51. Determine the actual value of the base current (I_B)
    a. 99.00 μA
    b. 990.0 μA
    c. 9.900 μA
    d. 0.9900 μA
    e. None of the above
52. Compare the values of I_B and I_2
    a. $I_2 \approx 20\ I_B$
    b. $I_2 \approx 40\ I_B$
    c. $I_2 \approx 10\ I_B$
    d. $I_2 \approx 30\ I_B$
    e. None of the above
53. Based on your comparison in **question 52** which of the following is a TRUE relationship between I_2 and I_B?
    a. $I_2 << I_B$
    b. $I_2 << 10I_B$
    c. $I_2 >> I_B$
    d. $I_2 >> 10I_B$
    e. None of the above
54. Given that the relationship you selected in **question 53** is TRUE, what will be the actual value of the voltage on the base of the transistor.
    a. 2.12V
    b. 2.00V
    c. 2.21V
    d. 2.20V
    e. None of the above
55. Determine the value of the collector current (I_C) at the Q-point.
    a. 1.12mA
    b. 1.00mA
    c. 1.21mA
    d. 1.20mA
    e. None of the above
56. Determine the value of the collector emitter voltage (V_{CE}) at the Q-point.
    a. 7.70V
    b. 6.78V
    c. 8.47V
    d. 5.47V
    e. None of the above
57. What is the value of (I_C) at saturation?
    a. 3.791mA
    b. 3.791μA
    c. 2.791μA
    d. 2.791mA
    e. None of the above

58. What is the value of (V_{CE}) at cut-off?
    a. 20.0V
    b. 9.98V
    c. 9.99V
    d. 12.0V
    e. None of the above

Assuming that the relationship you selected in **question 53** is **NOT TRUE** and the transistor in Figure. 1.10 has h_{FE} = 60 when at room temperature, T = 25 °C. Given that R_1 = 70kΩ, R_2 = 12kΩ, R_E = 1.3kΩ, R_C = 6.4kΩ, and V_{CC} = +22V.

Use figure 1.10 and the above information to answer questions from 59 to 69

59. Determine the value of the resistance the base of the transistor would experience
    a. 7.848K ohms
    b. 7848K ohms
    c. 78.48K ohms
    d. 784.8K ohms
    e. None of the above
60. Determine the effective resistance (R_{eq}) considering the input resistance (R_{IN}) and resistor (R_2)
    a. 10.4K ohms
    b. 11.4K ohms
    c. 12.4K ohms
    d. 13.4K ohms
    e. None of the above
61. Determine the value of the voltage drop across the effective resistance (R_{eq})
    a. 2.528V
    b. 2.484V
    c. 2.585V
    d. 2.848V
    e. None of the above
62. What will be the value of the voltage on the base of the transistor?
    a. 2.848V
    b. 2.484V
    c. 2.585V
    d. 2.528V
    e. None of the above
63. Determine the value of the voltage (V_E) on the emitter of the transistor
    a. 2.348V
    b. 2.484V
    c. 2.148V
    d. 2.228V
    e. None of the above

64. Determine the value of the emitter current (I_E).
    a. 1.653mA
    b. 1.653μA
    c. 2.653μA
    d. 2.653mA
    e. None of the above
65. Determine the value of the collector current (I_C) at the Q-point.
    a. 2.653μA
    b. 2.653mA
    c. 1.653mA
    d. 1.653μA
    e. None of the above
66. Determine the value of the collector emitter voltage (V_{CE}) at the Q-point.
    a. 9.57V
    b. 9.48V
    c. 9.39V
    d. 9.27V
    e. None of the above
67. What is the value of (I_C) at saturation?
    a. 1.37mA
    b. 1.37μA
    c. 2.73μA
    d. 2.73mA
    e. None of the above
68. What is the value of (V_{CE}) at cut-off?
    a. 23.00V
    b. 22.98V
    c. 21.99V
    d. 22.00V
    e. None of the above
69. Determine the value of the voltage drop across the resistor (R_C)
    a. 10.57V
    b. 10.48V
    c. 10.39V
    d. 10.27V
    e. None of the above
70. With the value of V_{RC} from question 60, determine the value of the current (I_C) flowing through R_C
    a. 2.652μA
    b. 2.652mA
    c. 1.652mA
    d. 1.652μA
    e. None of the above

71. Comparing the answers from question 65 and 70, we can conclude that the values are
    a. $I_2 \approx I_B$
    b. Not related
    c. Different
    d. Equal
    e. None of the above
72. Which of the following can be considered as a(n) disadvantage of the biasing type shown in figure 1.10
    a. Requires more capacitors than most other biasing circuits
    b. Requires more components than most all other biasing circuits
    c. Requires more components than most fixed bias circuits
    d. Requires more components than most emitter bias circuits
    e. None of the above

Preamble for questions (73 to 75) Given that a transistor has a $\beta = 150$ and $I_E = 7.00$mA

73. What is the value of I_B?
    a. 4.636μA
    b. 46.60μA
    c. 46.36μA
    d. 463.6μA
    e. None of the above
74. What is the value of I_C?
    a. 6.954mA
    b. 6.095mA
    c. 695.4mA
    d. 69.54mA
    e. None of the above
75. Find the value of the ratio I_B: I_C.
    a. 1:21
    b. 1:32
    c. 1:42
    d. 1:52
    e. None of the above
76. After studying basic electronics at KNUST, John Dumelo decides to build a microphone amplifier. He chooses to use an NPN transistor in common emitter configuration and then connects it to his power supply. John Dumelo then connects the microphone directly to the base and a speaker directly to the output. When John Dumelo talks into the amplifier, he hears nothing. What should John Dumelo do to improve his circuit?
    a. He should buy a different microphone
    b. John Dumelo should buy a different speaker
    c. John Dumelo should increase the power supply voltage
    d. John Dumelo should apply proper biasing to the transistor
    e. None of the above

77. In BJT's "Beta" is sometimes referred to as .... gain
    a. Voltage
    b. Current
    c. Transconductance
    d. Resistive
    e. None of the above
78. DJ Kalid wants to design an amplifier to be used to play music in his crib. DJ Kalid is in final year and for some reason he has forgotten a lot of his basic electronics, hence he isn't sure of what biasing scheme to use. Which biasing scheme will you recommend to DJ Kalid as an engineer in the making
    a. Common-Emitter Biasing
    b. Common-Base Biasing
    c. Common-Collector Biasing
    d. All of the above
    e. None of the above
79. Which of the following statements is true about an NPN transistor in the active region?
    a. The Collector-Base junction is reverse biased
    b. The Base-Emitter junction is reverse biased
    c. The Collector-Base junction is forward biased
    d. All of the above
    e. None of the above
80. Which of the following statements is true about an NPN transistor in saturation?
    a. Base-Emitter junction is reverse biased
    b. Collector-Base junction is reverse biased
    c. Collector-Base junction is foward biased
    d. All of the above
    e. None of the above
81. Which of the following statements is true about an NPN transistor in cutt-off?
    a. Base-Emitter junction is foward biased
    b. There is no current flow at the collector
    c. Collector-Base junction is foward biased
    d. All of the above
    e. None of the above
82. On the datasheet of transistors β_{dc} (Beta dc) can sometimes be represented as?
    a. H_{XE}
    b. H_{LE}
    c. H_{YE}
    d. H_{FE}
    e. None of the above
83. Kwamena is designing a common-emitter BJT amplifier, which of the following terminals of the BJT can Kwamena connect his speaker to?
    a. Base
    b. Collector

c. Emitter
d. All of the above terminals
e. None of the above

84. Kwaku wants to design a common-base BJT amplifier, which of the following terminals would you advice him to connect his microphone to?
    a. Base
    b. Collector
    c. Emitter
    d. All of the above terminals
    e. None of the above
85. In bipolar junction transistors, why is the base very thin?
    a. To allow the free flow of electrons
    b. To allow carriers injected into the base to diffuse into the collector
    c. To prevent carriers injected into the base to diffuse into the collector
    d. None of the above
86. Which of the following statements about BJT's is true?
    a. BJT's have only one mode of operation
    b. The currents flowing through the BJT are temperature dependent
    c. The currents flowing through the BJT are temperature independent
    d. At room temperature VT = 0.7V
    e. None of the above
87. The transconductance (gm) of a transistor relates which two quantities?
    a. Change in Emitter current and change in Collector current
    b. Change in Base current and change in Collector current
    c. Change in Collector current and change in Base-Emitter voltage
    d. Change in Base current and change in Base-Emitter voltage
    e. None of the above

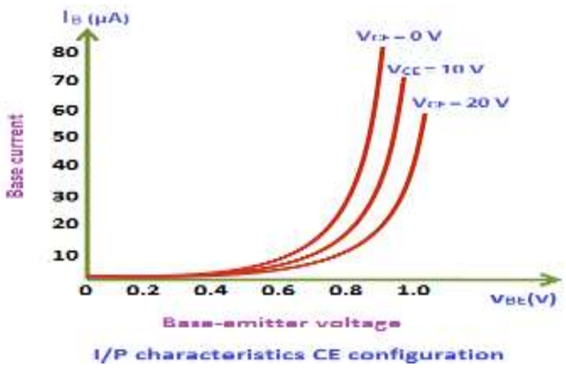

Figure 1.11

Figure 1.11 represents the input characteristics of a transistor in the common emitter configuration. Using figure 1.11, answer the **questions 88 to 92**

 G. S. KLOGO

88. Given that the turn on voltage of the transistor is 0.8V, what is the approximate value of I_B, when the output voltage is kept constant at 0V
    a. 10.00 mA
    b. 20.00 μA
    c. 10.00 μA
    d. 20.00 mA
    e. None of the above
89. Given that the turn on voltage of the transistor is 0.8V, what is the approximate value of I_B, when the output voltage is kept constant at 10V
    a. 10.00 mA
    b. 20.00 μA
    c. 10.00 μA
    d. 20.00 mA
    e. None of the above
90. Given that the turn on voltage of the transistor is 0.8V, what is the approximate value of I_B, when the output voltage is kept constant at 20V
    a. 10.00 μA
    b. 30.00 μA
    c. 40.00 μA
    d. 20.00 μA
    e. None of the above
91. What is the relationship between I_B and V_{CE}
    a. Increasing V_{CE} results in increase in I_B
    b. Increasing V_{CE} results in decrease in I_B
    c. decreasing I_B results in increase in V_{BE}
    d. Increasing V_{BE} results in decrease in I_B
    e. None of the above
92. What is the significance of the relationship in **question 88**
    a. Increasing V_{CB} results in increase in I_B
    b. Increasing V_{EE} results in decrease in I_B
    c. decreasing I_B results in increase in V_{BE}
    d. Increasing V_{BE} results in decrease in I_B
    e. None of the above
93. In BJT amplifier design the active region is sometimes referred to as the......................
    a. Activated Region
    b. "On" Region
    c. Linear Region
    d. "Off" Region
94. Which of the following statements about derating BJT's is true? (PDmax = Maximum Power Dissipation)
    a. PDmax decreases with increasing temperature
    b. PDmax is insensitive to temperature change
    c. PDmax increases with increasing temperature
    d. PDmax decreases with decreasing temperature